AI-Generated Text Detection

A Comparative Study of Classical, Recurrent, and Transformer-Based Approaches

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The Problem

Why AI text detection matters

Education: academic dishonesty in essays and assignments

Journalism: fabricated quotes and sources in news articles

Moderation: coordinated inauthentic behavior on platforms

Research integrity: AI-authored papers undermining science

ChatGPT

100M+

active users in 2 months

Faster adoption than any consumer technology in history.

Our goal: *Build a binary classifier that distinguishes human writing from ChatGPT output, then audit it for fairness.*

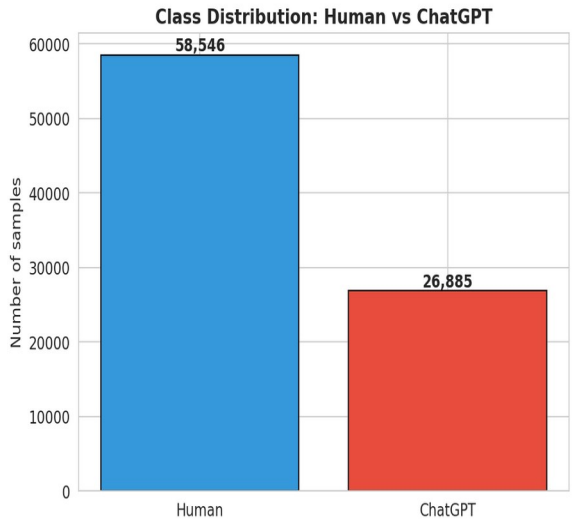
The Dataset: HC3

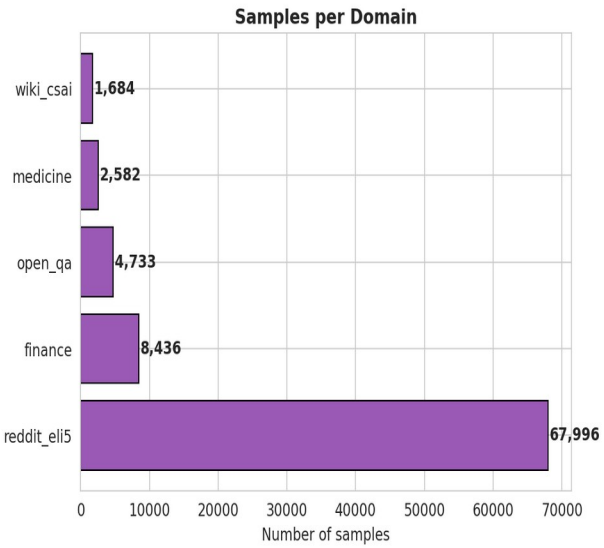
Human ChatGPT Comparison Corpus (Guo et al., 2023)

~85K	5	2	EN
text samples	domains	classes (Human / AI)	language

Domain breakdown (test set)

- Reddit ELI5 — 8,727 samples (casual)
- Finance — 1,240 samples
- Open-domain Q&A — 652 samples
- Medicine — 378 samples
- Wiki CSAI — 241 samples (technical)





What the Data Told Us

Three findings before we even built a model

Length

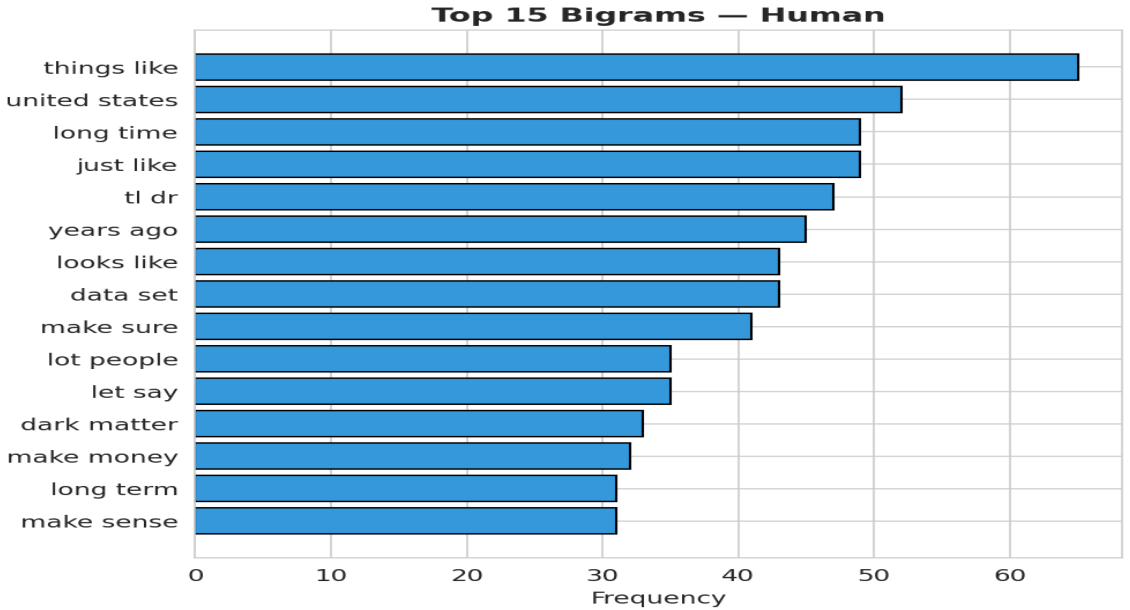
ChatGPT writes longer and more uniform responses than humans.

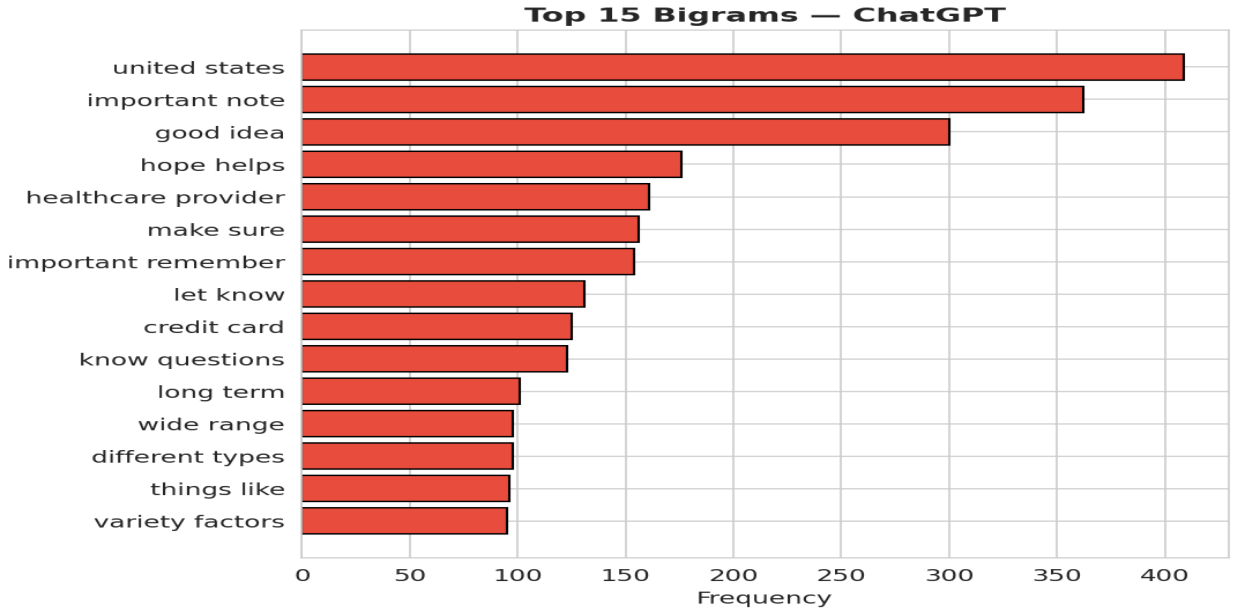
Vocabulary

AI overuses transitions: "however", "important", "additionally"

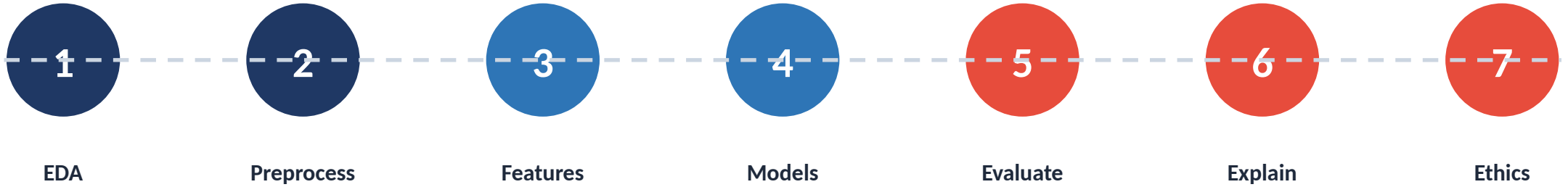
References

Humans frequently include URLs and links — ChatGPT does not.





End-to-End Pipeline



Models compared

3 classical: Logistic Regression, Linear SVM, Naive Bayes (TF-IDF features)

1 recurrent: Bi-LSTM with frozen GloVe-100d embeddings

1 transformer: DistilBERT fine-tuned (binary head)

Reproducibility

- All random seeds fixed at 42
- Stratified 70/15/15 train/val/test split
- Public dataset (Hugging Face HC3)
- Saved models, requirements.txt, README

Results: All 5 Models

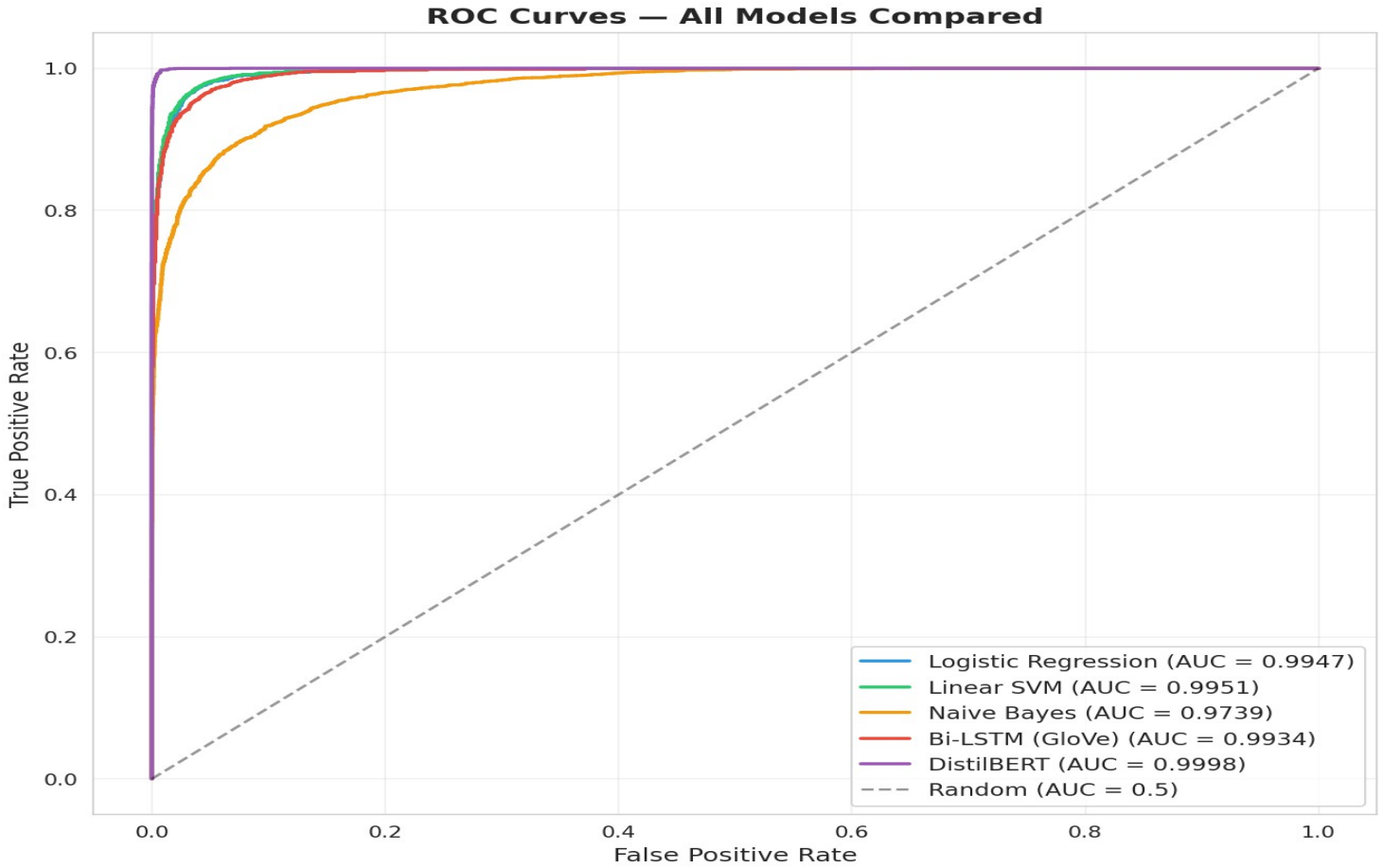
DistilBERT wins, but Linear SVM is surprisingly close

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.9666	0.9508	0.9539	0.9523	0.9947
Linear SVM	0.9672	0.9513	0.9549	0.9531	0.9951
Multinomial Naive Bayes	0.9169	0.9362	0.8179	0.8731	0.9739
Bi-LSTM (GloVe)	0.9518	0.8968	0.9740	0.9338	0.9934
DistilBERT 🏆	**0.9891**	**0.9706**	**0.9992**	**0.9847**	**0.9998**

Key insight: *DistilBERT's gain over Linear SVM is statistically significant (McNemar's $\chi^2 = 144.0$, $p = 3.55e\text{-}33$) — but only ~3 F1 points. Classical models remain a strong, fast, interpretable baseline.*

ROC: Near-Perfect Ranking by All Models

Every model achieves AUC > 0.97 — distinguishing human from AI is easy in aggregate.



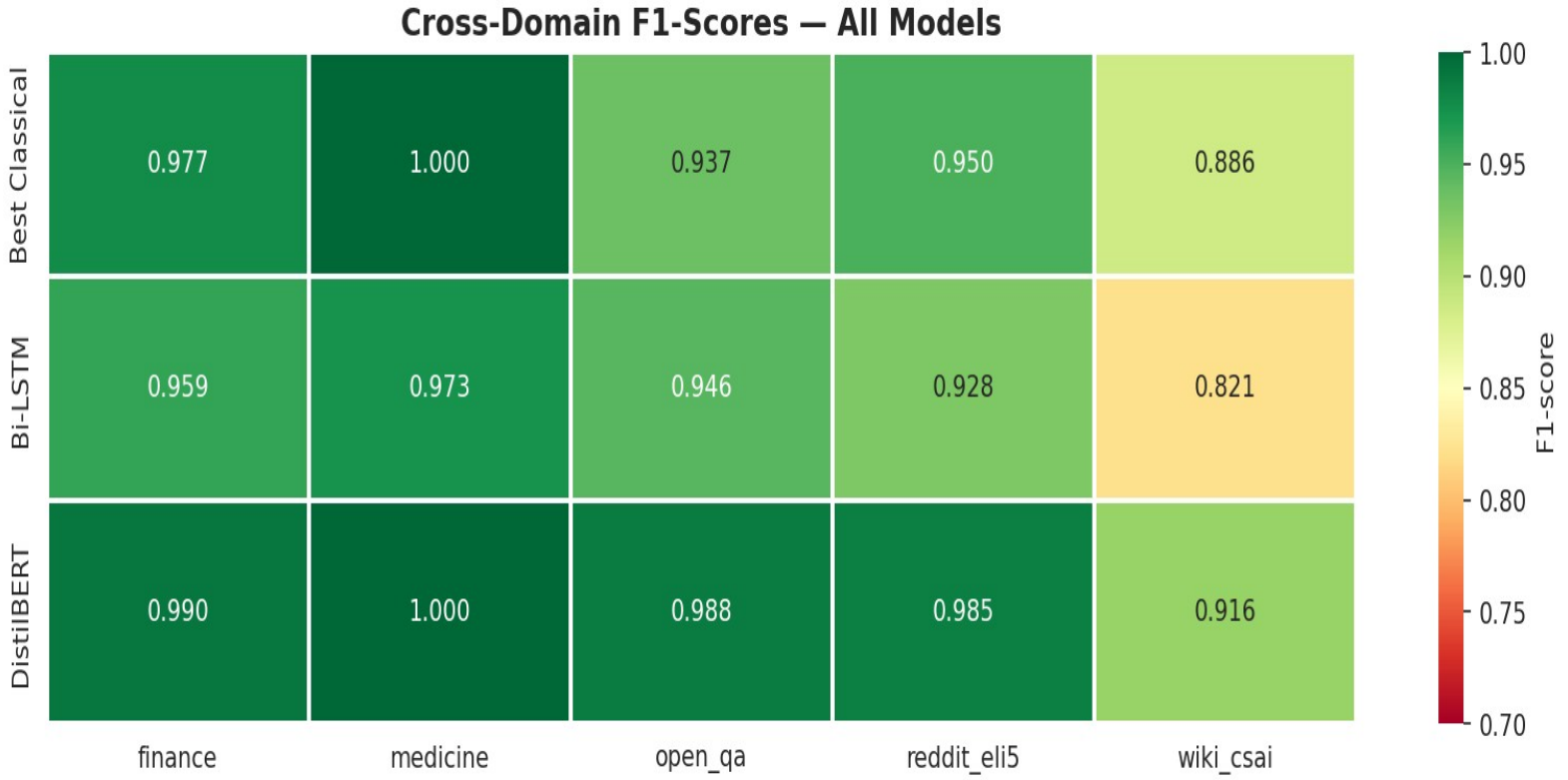
DistilBERT
AUC = 0.9998

Bi-LSTM
AUC = 0.9934

Linear SVM
AUC = 0.9951

Aggregate Metrics Hide Critical Disparities

F1 varies dramatically by domain — a single number tells only part of the story



The story

Easiest: medicine
DistilBERT F1 = 1.000
0% error rate (0/378)

Hardest: wiki_csai
DistilBERT F1 = 0.916
9.54% error rate (23/241)

Same pattern across LSTM and classical models — *this is a property of the data, not any single model.*

What the Models Actually Learn

Linguistic fingerprints surface immediately when we open the black box

🤖 Strong AI indicators

- **important**
- **including**
- **help**
- **even though**
- **additionally**

Formal transitions, qualifying language, structured prose

👤 Strong human indicators

- **url**
- **etc**
- **basically**
- **thus**
- **pretty**

Informal connectives, abbreviations, links to external sources

Ethics: The 9.54% Problem

Domain bias has real-world fairness consequences

Error rate

on wiki_csai

9.54%

23 out of 241 technical writers would be misclassified by the model

Non-native speakers

Liang et al., 2023 (Patterns)

Non-native English: **>50% misclassified as AI**

Native English: **<5% misclassified as AI**

Why? *Non-native writing has lower lexical diversity and more formal phrasing — exactly the features detectors flag as AI.*

⚠️ **International students would be disproportionately accused.**

Responsible Deployment

If we must deploy these systems, eight non-negotiable safeguards

1 Never sole evidence

Always require human review

2 Confidence intervals

Show P(AI), not binary labels

3 Domain calibration

Test in target context

4 Fairness audit

Test diverse populations

5 Avoid punitive use

Education, hiring carry stakes

6 Continuous re-training

Every 3-6 months

7 Transparency

Right to challenge results

8 Don't build it?

Sometimes the right call

Limitations & Future Work

⚠️ Limitations

- Only ChatGPT (gpt-3.5) tested — no GPT-4 or Gemini
- English only — multilingual untested
- 5 domains — narrow slice of real text
- No adversarial testing (paraphrasing attacks)
- Static evaluation — performance degrades over time

🚀 Future Work

- Robustness against newer LLMs (GPT-4, Gemini)
- Multilingual extension (Arabic, Spanish, French)
- Adversarial training against paraphrasing
- Fairness-aware loss functions
- Real-time deployment with confidence intervals

Thank You

Questions & Discussion

5 models • **F1 = 0.9847** • **AUC = 0.9998**

Statistical significance + Cross-domain analysis + Bias audit